Experiment 5: Out of Boundary Paths

Aim: To find the number of paths by which a ball can move out of the boundary of a grid within a given maximum number of moves.

Algorithm

1. Start the program.
2. Read the number of rows, columns, maximum moves, and starting position.
3. Initialize a DP array with zero and set the starting cell to one path.
4. For each move, examine every cell containing a possible path.
5. If moving in a direction takes the ball outside the grid, add the number of paths to the answer.
6. Otherwise, add the paths to the corresponding neighboring cell.
7. Repeat until the maximum number of moves is reached.
8. Display the total number of paths modulo \(10^9+7\).
9. Stop the program.

C

```
#include <stdio.h>

#define MOD 1000000007

int main() {

    int m, n, maxMove, startRow, startColumn;

    int dp[51][51] = {0};

    int next[51][51];

    int moves, i, j;

    long long paths = 0;

    printf("Enter rows and columns: ");

    scanf("%d %d", &m, &n);

    printf("Enter maximum moves: ");

    scanf("%d", &maxMove);

    printf("Enter starting row and column: ");

    scanf("%d %d", &startRow, &startColumn);

    dp[startRow][startColumn] = 1;


    for (moves = 1; moves <= maxMove; moves++) {
```

```
for (i = 0; i < m; i++)
    for (j = 0; j < n; j++)
        next[i][j] = 0;
for (i = 0; i < m; i++) {
    for (j = 0; j < n; j++) {
        if (dp[i][j] == 0)
            continue;
        int count = dp[i][j];
        if (i == 0)
            paths = (paths + count) % MOD;
        else
            next[i - 1][j] =
            (next[i - 1][j] + count) % MOD;
        if (i == m - 1)
            paths = (paths + count) % MOD;
        else
            next[i + 1][j] =
            (next[i + 1][j] + count) % MOD;
        if (j == 0)
            paths = (paths + count) % MOD;
        else
            next[i][j - 1] =
            (next[i][j - 1] + count) % MOD;
        if (j == n - 1)
            paths = (paths + count) % MOD;
        else
            next[i][j + 1] =
            (next[i][j + 1] + count) % MOD;
    }
```

```
        }
        for (i = 0; i < m; i++)
            for (j = 0; j < n; j++)
                dp[i][j] = next[i][j];
    }
    printf("Number of paths: %lld\n", paths);
    return 0;
}
```

Input:

Enter number of rows: 2

Enter number of columns: 2

Enter maximum moves: 2

Enter starting row: 0

Enter starting column: 0

Output:

```
Enter rows and columns: 2
2
Enter maximum moves: 2
Enter starting row and column: 0
0
Number of paths: 6
```

Result

Thus, the C program successfully calculates the number of paths by which a ball can move out of the grid boundary within the given maximum number of moves.